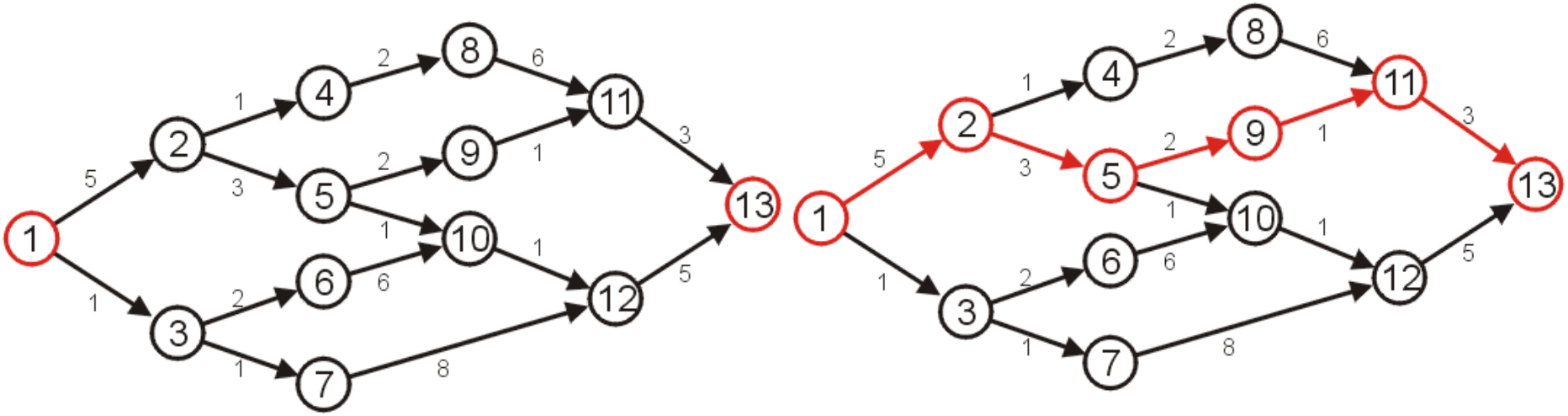
Data Structures

17. Djikstra's Algorithm

Shortest Path

- Given a weighted graph
  - Problem is to find the shortest path between two given vertices
- Length of a path in a weighted graph
  - Sum of the weights of each of the edges in that path
- Example: Shortest path from vertex 1 to vertex 13
  - Other paths exists but they are longer

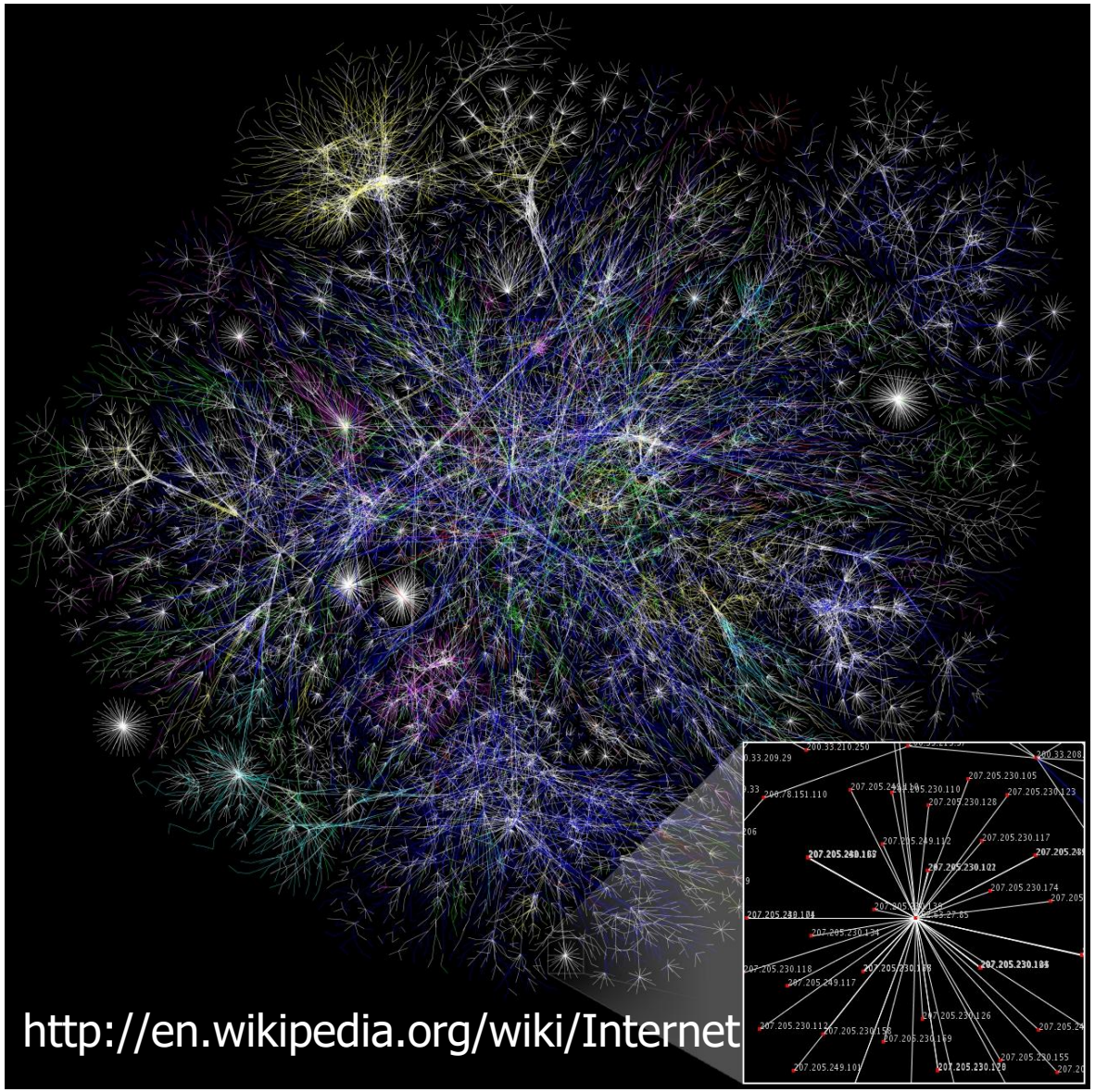


Application – Computer Networks

- The Internet is a collection of interconnected computer networks
  - Information is passed through packets
- Packets are passed from the source, through routers, to their destination
- Routers are connected to either:
  - Individual computers, or
  - Other routers
- These may be represented as graphs

Application – Computer Networks

- A visualization of the graph of the routers and their various connections through a portion of the Internet



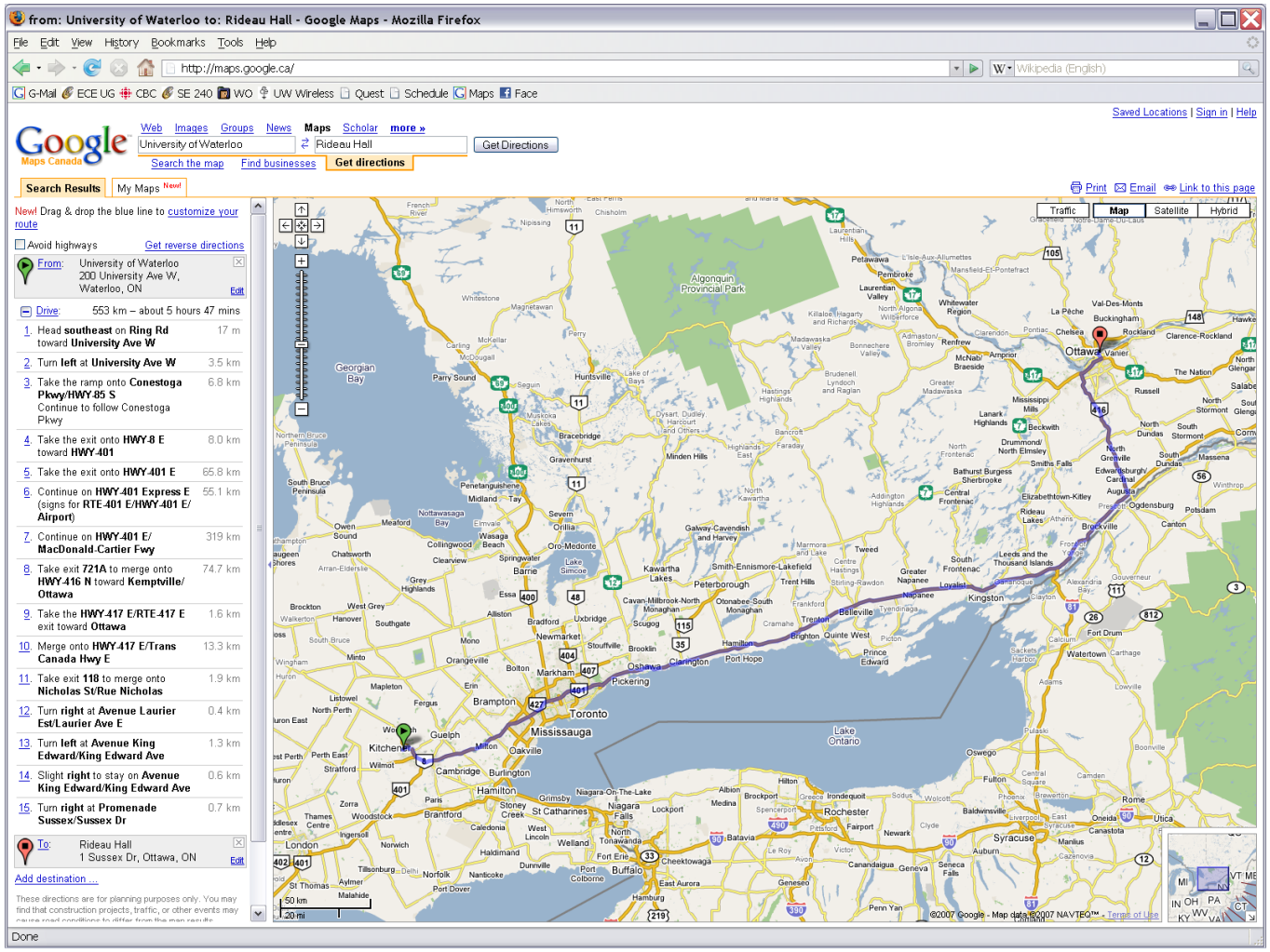
Application – Computer Networks

- The path a packet takes depends on the IP address

- Metrics for measuring the shortest path may include
  - Low latency (minimize time)
  - Minimum hop count (all edges have weight 1)

Application – Traffic

- Find the shortest route between to points on a map
  - Shortest path, however, need not refer to distance…



Variants of Shortest Path

Given a graph `G = (V, E)`

- Single-source shortest paths
  - Find shortest path from a given source vertex s to each vertex $v \in V$
- Single-destination shortest paths
  - Find shortest path to a given destination vertex t from each vertex v
- Single-pair shortest path
  - Find shortest path from u to v for given vertices u and v
- All-pairs shortest-paths
  - Find shortest path from u to v for every pair of vertices u and v

Single Source Shortest Path

Dijkstra's Algorithm

- Problem: From a given source vertex $s \in V$, find the shortest-paths and their weights $w(s,v)$ for all $v \in V$

- Idea of the Algorithm
  - Maintain a set S of vertices whose shortest-path distances from s are known
  - At each step add to S the vertex $v \in V-S$ whose distance estimate from s is minimal
  - Update the distance estimates of vertices adjacent to v

Dijkstra's Algorithm - Pseudocode

```
dist[s] = 0                          // distance to source vertex is zero
p[s] = NULL
for all v ∈ V–{s}
   dist[v] = ∞                       // set all other distances to infinity
S = ∅                                //S, the set of visited vertices is initially empty
Q = V                                //Q, the queue initially contains all vertices
while Q is not emppty                //while the queue is not empty
   u =  mindistance(Q)               //select the element of Q with the min distance
   S = S∪{u}                         // add u to list of visited vertices
   for all v ∈ neighbors[u]
      if dist[v] > dist[u] + w(u,v)   //if new shortest path found
         dist[v] = dist[u] + w(u,v)   //set new value of shortest path
         p[v] = u
```

Dijkstra's Vs. Prim's Algorithm

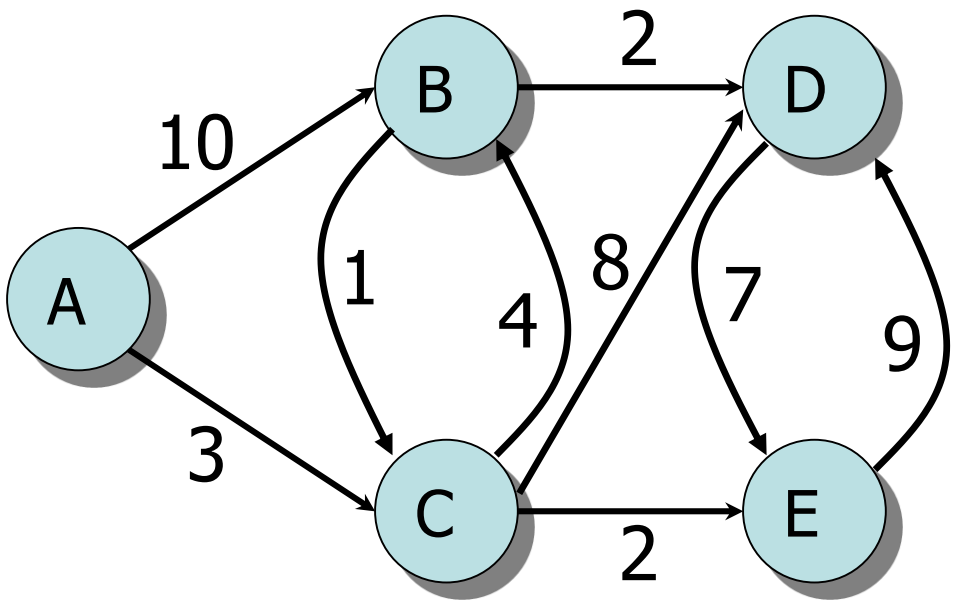
Dijkstra's Algorithm

```
dist[s] = 0
p[s] = NULL
for all v ∈ V–{s}
   dist[v] = ∞
S = ∅
Q = V
while (Q not empty)
   u =  mindistance(Q)
   S = S∪{u}
   for all v ∈ neighbors[u] && v∈V-S
      if dist[u] + w(u, v) < dist[v]
         dist[v] = dist[u] + w(u, v)
         p[v] = u
```

Prim's Algorithm

```
Q = V[G];
for each u ∈ Q
   key[u] = ∞;
key[r] = 0;
p[r] = NULL;
while (Q not empty)
   u = ExtractMin(Q);
   for each v ∈ Adj[u]
      if (v ∈ Q and w(u,v) < key[v])
         p[v] = u;
         key[v] = w(u,v);
```

Dijkstra's Algorithm – Example



Q:

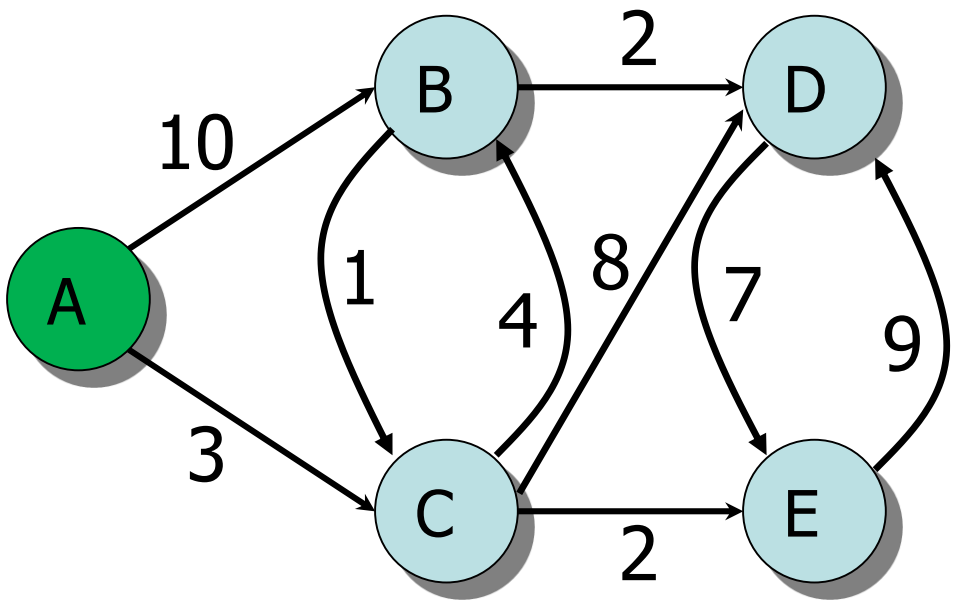
A	B	C	D	E
0	∞	∞	∞	∞

Initialization

S: {}

Vertex	Distance	Parent
A	0	Ø
B	∞	Ø
C	∞	Ø
D	∞	Ø
E	∞	Ø

Dijkstra's Algorithm – Example



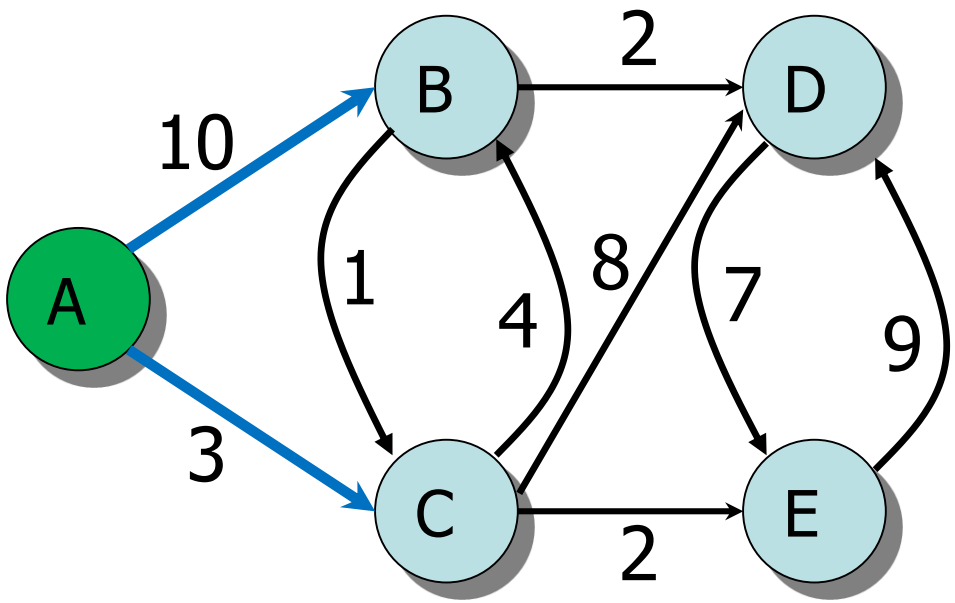
Q:	A	B	C	D	E
	0	∞	∞	∞	∞

A ← EXTRACT-MIN(Q)

S: {A}

Vertex	Distance	Parent
A	0	Ø
B	∞	Ø
C	∞	Ø
D	∞	Ø
E	∞	Ø

Dijkstra's Algorithm – Example



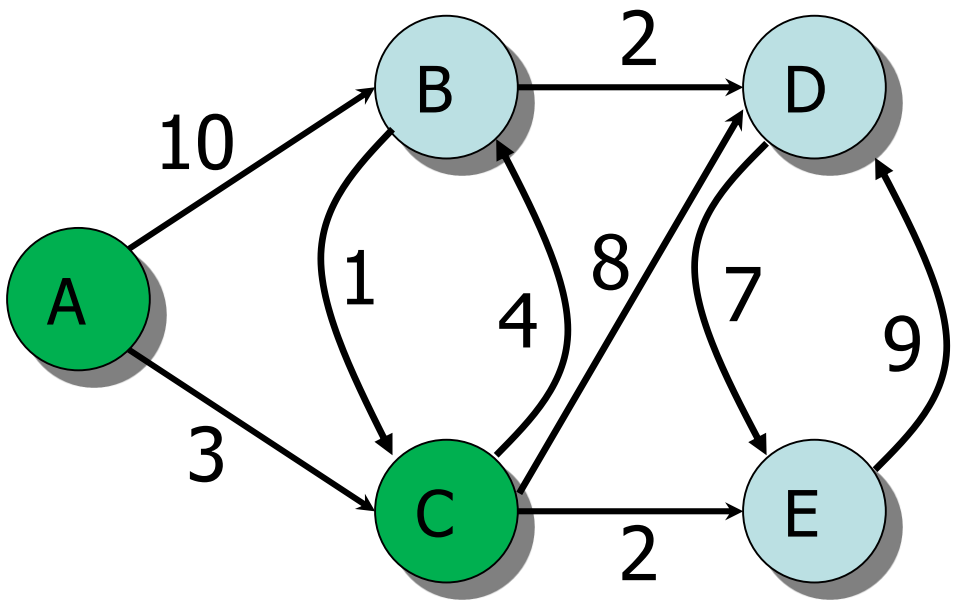
Q:	A	B	C	D	E
	0	∞	∞	∞	∞
		10	3	∞	∞

Update all neighbors of A

S: {A}

Vertex	Distance	Parent
A	0	Ø
B	10	A
C	3	A
D	∞	Ø
E	∞	Ø

Dijkstra's Algorithm – Example



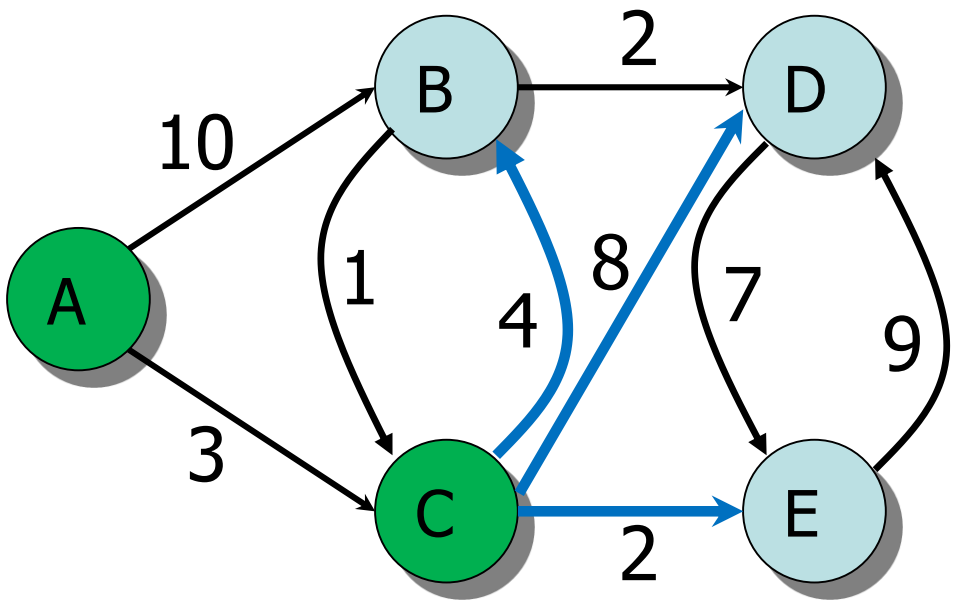
Q:	A	B	C	D	E
	0	∞	∞	∞	∞
		10	3	∞	∞

C ← EXTRACT-MIN(Q)

S: {A, C}

Vertex	Distance	Parent
A	0	Ø
B	10	A
C	3	A
D	∞	Ø
E	∞	Ø

Dijkstra's Algorithm – Example



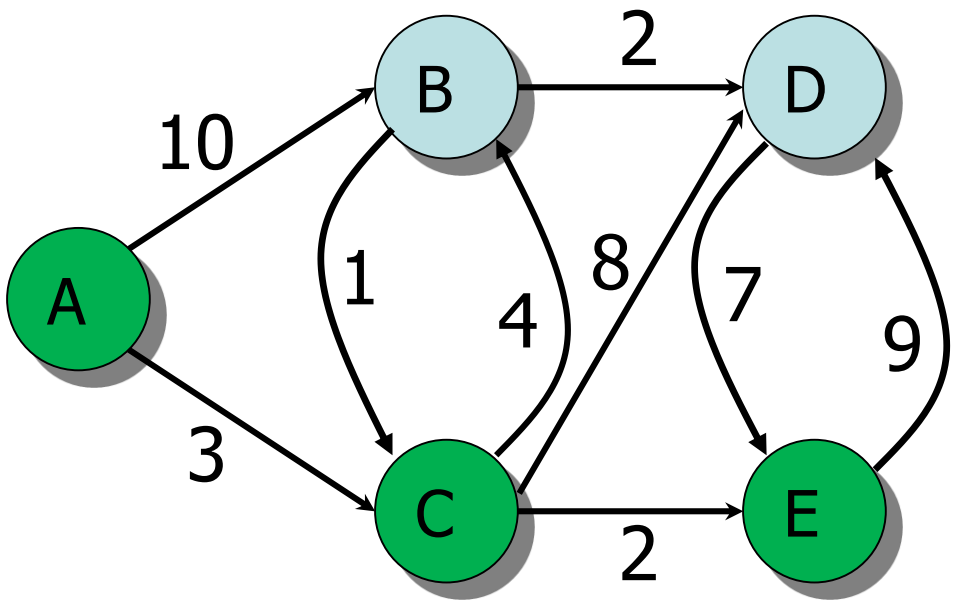
Q:	A	B	C	D	E
	0	∞	∞	∞	∞
		10	3	∞	∞
		7		11	5

Update all neighbors of C

S: {A, C}

Vertex	Distance	Parent
A	0	Ø
B	7	C
C	3	A
D	11	C
E	5	C

Dijkstra's Algorithm – Example



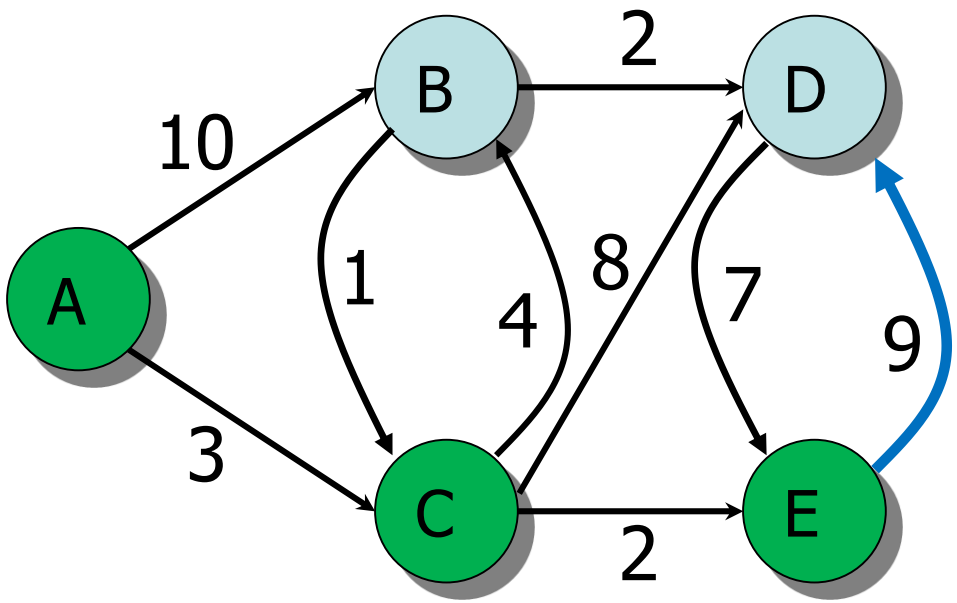
Q:	A	B	C	D	E
	0	∞	∞	∞	∞
		10	3	∞	∞
		7		11	5

E ← EXTRACT-MIN(Q)

S: {A, C, E}

Vertex	Distance	Parent
A	0	Ø
B	7	C
C	3	A
D	11	C
E	5	C

Dijkstra's Algorithm – Example



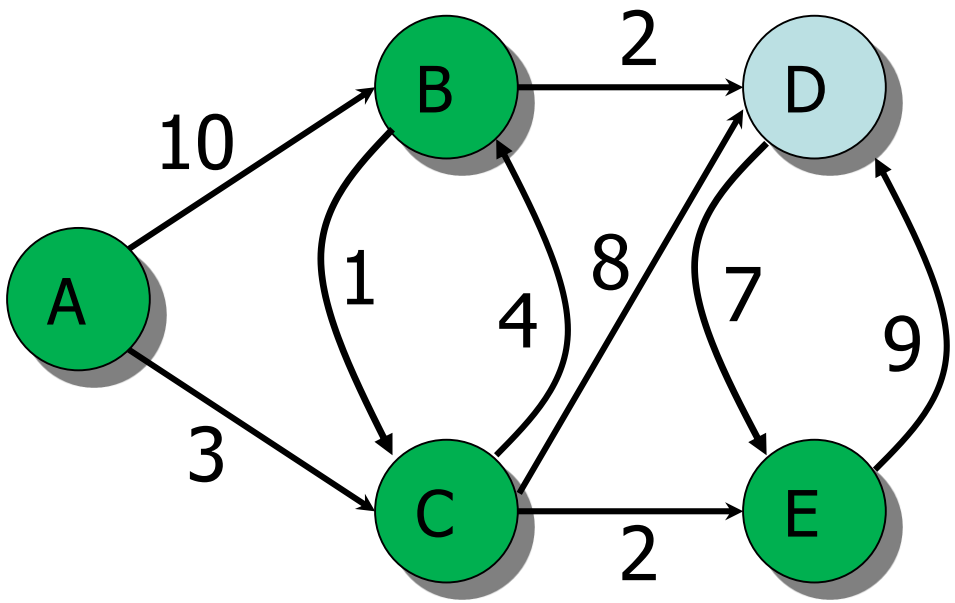
Q:	A	B	C	D	E
	0	∞	∞	∞	∞
		10	3	∞	∞
		7		11	5
		7		11	

Update all neighbors of E

S: {A, C, E}

Vertex	Distance	Parent
A	0	Ø
B	7	C
C	3	A
D	11	C
E	5	C

Dijkstra's Algorithm – Example



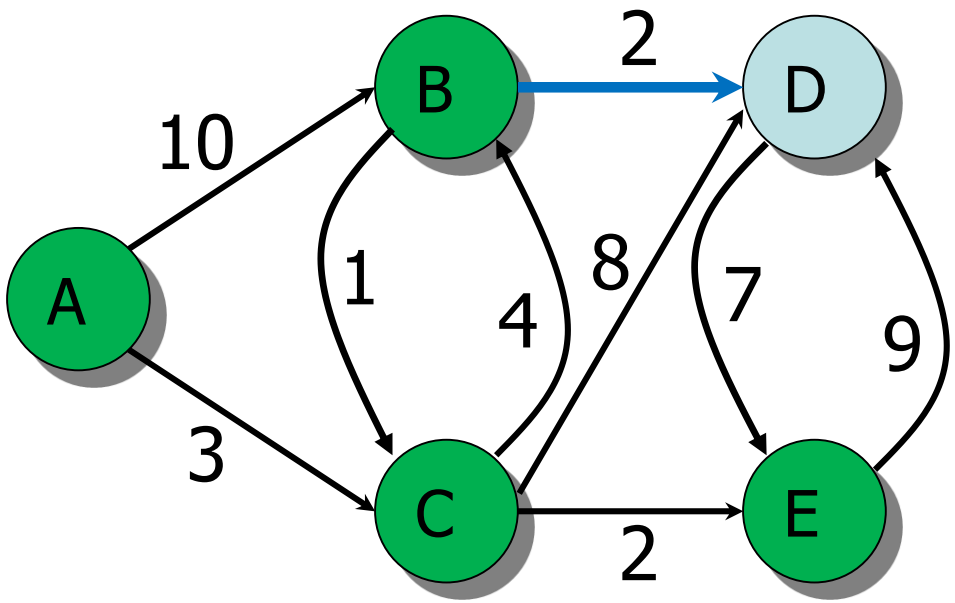
Q:	A	B	C	D	E
	0	∞	∞	∞	∞
		10	3	∞	∞
		7		11	5
		7		11	

B ← EXTRACT-MIN(Q)

S: {A, C, E, B}

Vertex	Distance	Parent
A	0	Ø
B	7	C
C	3	A
D	11	C
E	5	C

Dijkstra's Algorithm – Example



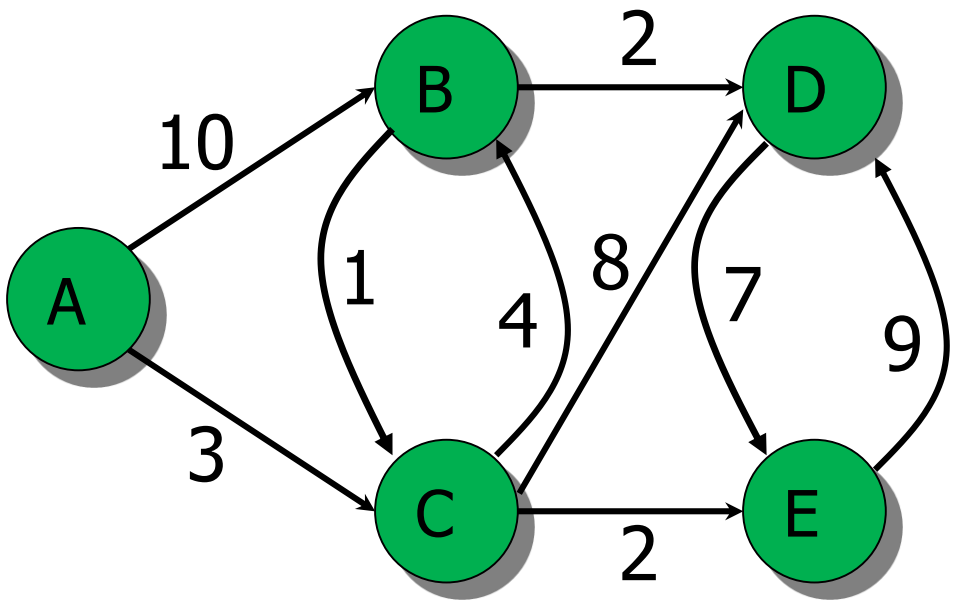
Q:	A	B	C	D	E
	0	∞	∞	∞	∞
		10	3	∞	∞
		7		11	5
		7		11	
				9	

Update all neighbors of B

S: {A, C, E, B}

Vertex	Distance	Parent
A	0	Ø
B	7	C
C	3	A
D	9	B
E	5	C

Dijkstra's Algorithm – Example



Q:	A	B	C	D	E
	0	∞	∞	∞	∞
		10	3	∞	∞
		7		11	5
		7		11	
				9	

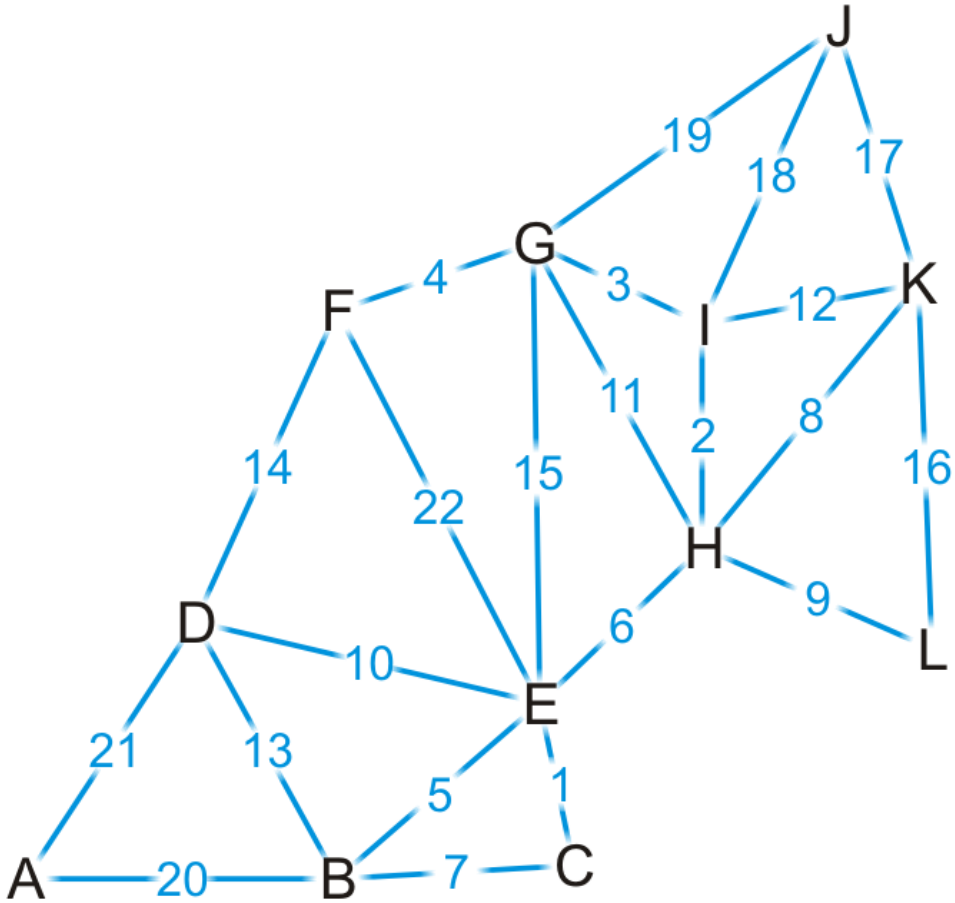
D ← EXTRACT-MIN(Q)

S: {A, C, E, B, D}

Vertex	Distance	Parent
A	0	Ø
B	7	C
C	3	A
D	9	B
E	5	C

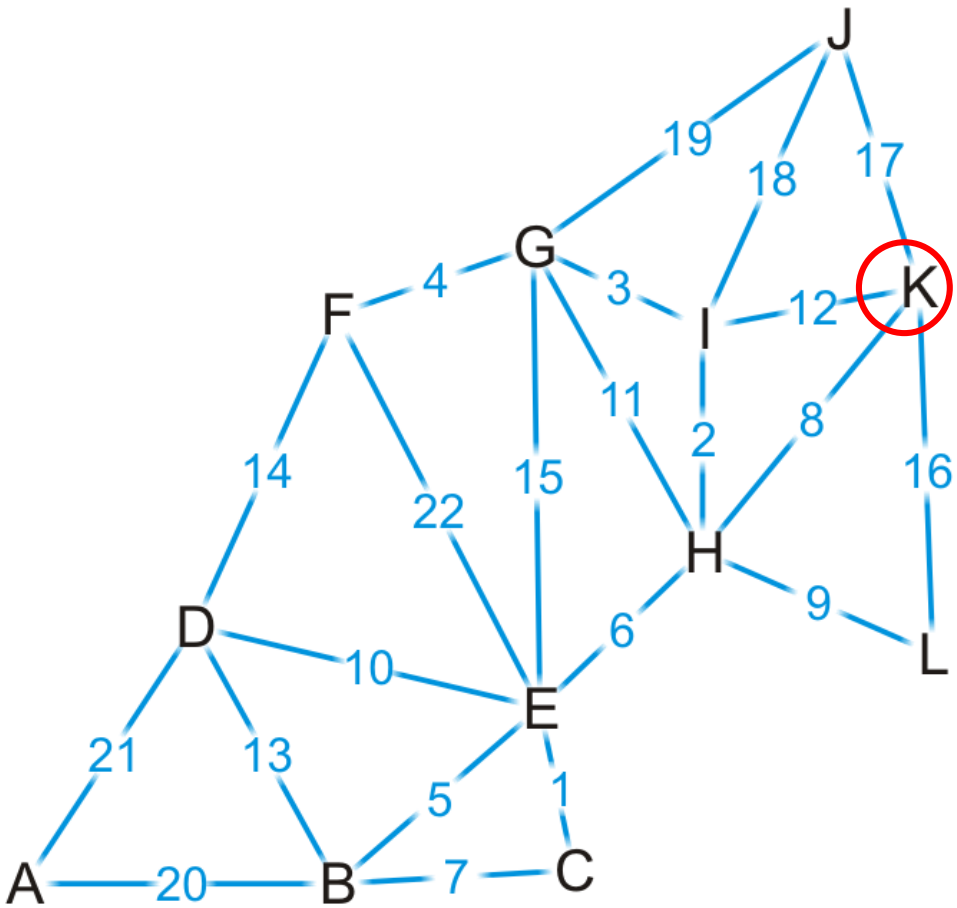
Dijkstra's Algorithm – Example

- Find the shortest path from K to every other vertex



Dijkstra's Algorithm – Example

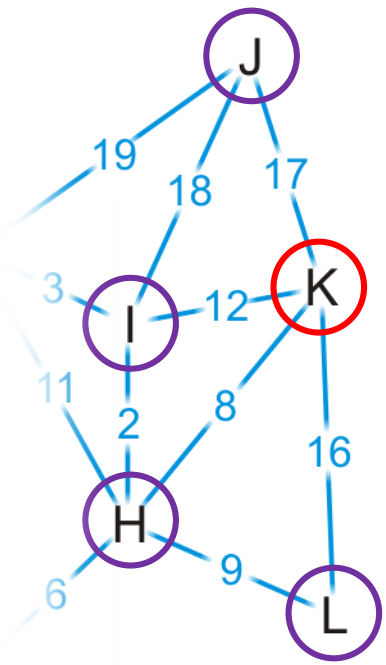
- We visit vertex K



Vertex	S	Distance	Parent
A	F	∞	Ø
B	F	∞	Ø
C	F	∞	Ø
D	F	∞	Ø
E	F	∞	Ø
F	F	∞	Ø
G	F	∞	Ø
H	F	∞	Ø
I	F	∞	Ø
J	F	∞	Ø
K	**T**	**0**	**Ø**
L	F	∞	Ø

Dijkstra's Algorithm – Example

- Vertex K has four neighbors: H, I, J and L



Vertex	S	Distance	Parent
A	F	∞	Ø
B	F	∞	Ø
C	F	∞	Ø
D	F	∞	Ø
E	F	∞	Ø
F	F	∞	Ø
G	F	∞	Ø
H	**F**	**∞**	**Ø**
I	**F**	**∞**	**Ø**
J	**F**	**∞**	**Ø**
K	**T**	**0**	**Ø**
L	**F**	**∞**	**Ø**

Comments on Dijkstra's Algorithm

- If at some point, all unvisited vertices have a distance ∞?
  - This means that the graph is unconnected
  - We have found the shortest paths to all vertices in the connected subgraph containing the source vertex

- To find the shortest path between vertices v_j and v_k?
  - Apply the same algorithm, but stop when visiting vertex v_k

- Does the algorithm change if graph is directed?
  - No

Any Question So Far?

Graph Theory

Shortest Paths II

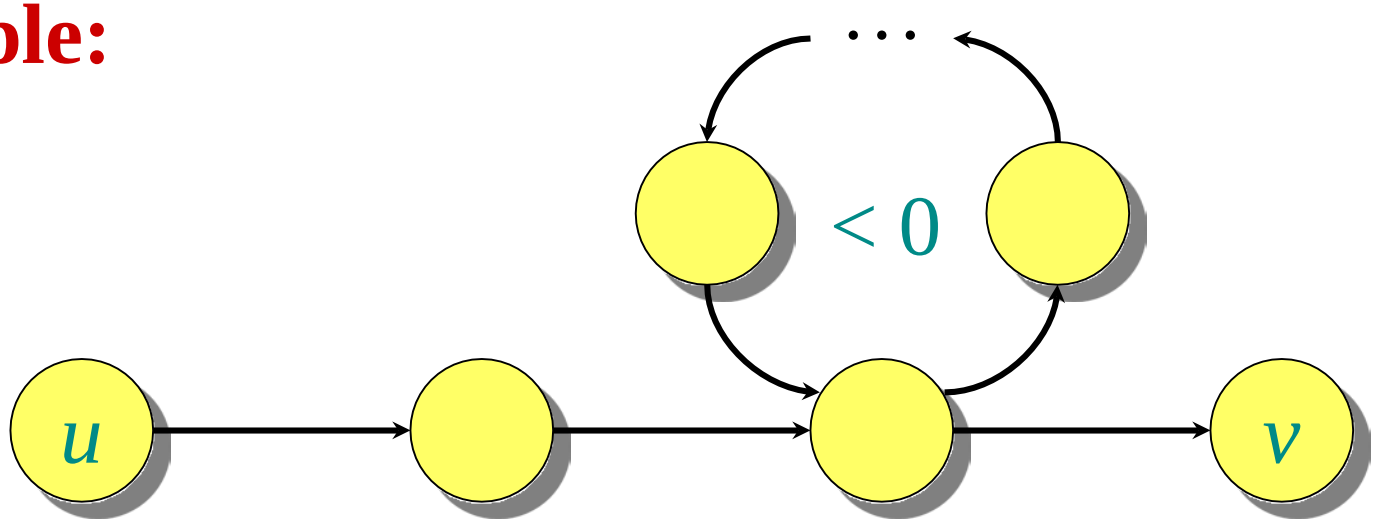
Bellman-Ford algorithm

Design and Analysis of Algorithms

Negative-weight cycles

Recall: If a graph $G = (V, E)$ contains a negative-weight cycle, then some shortest paths may not exist.

Example:



Bellman-Ford algorithm: Finds all shortest-path lengths from a ***source*** $s \in V$ to all $v \in V$ or determines that a negative-weight cycle exists.

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Bellman-Ford algorithm

$d[s] \leftarrow 0$
for each $v \in V - \{s\}$
  do $d[v] \leftarrow \infty$

} initialization

for $i \leftarrow 1$ **to** $|V| - 1$
  do for each edge $(u, v) \in E$
    do if $d[v] > d[u] + w(u, v)$
      then $d[v] \leftarrow d[u] + w(u, v)$

} ***relaxation step***

for each edge $(u, v) \in E$
  do if $d[v] > d[u] + w(u, v)$
    then report that a negative-weight cycle exists

At the end, $d[v] = \delta(s, v)$, if no negative-weight cycles.
Time $= O(VE)$.

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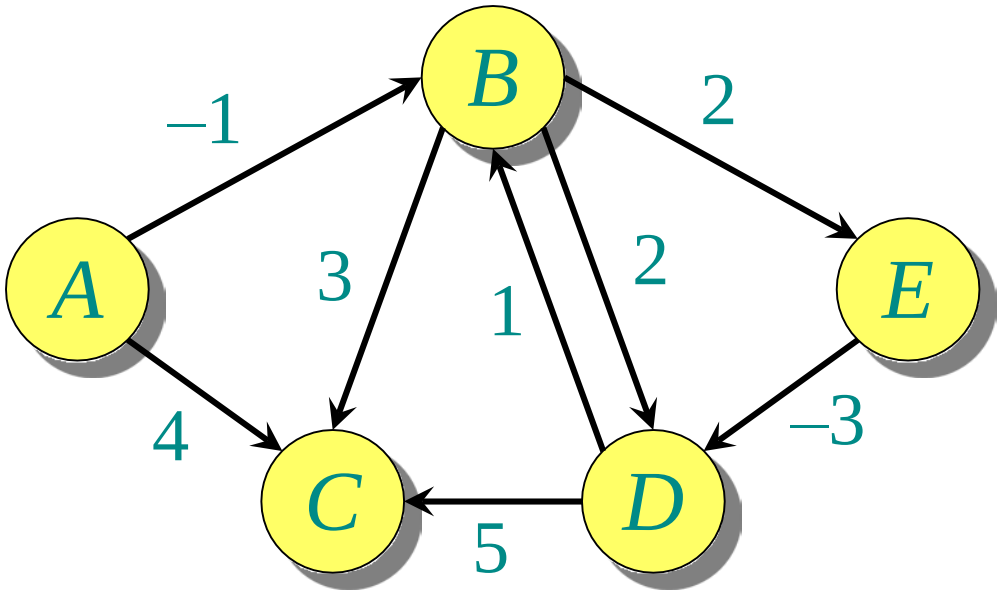
Example of Bellman-Ford



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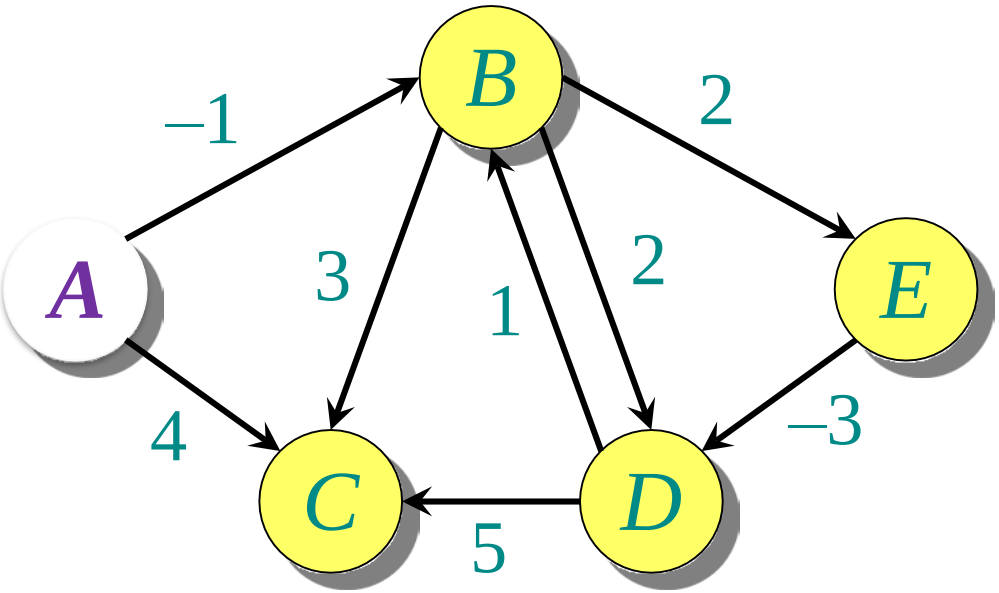
Example of Bellman-Ford



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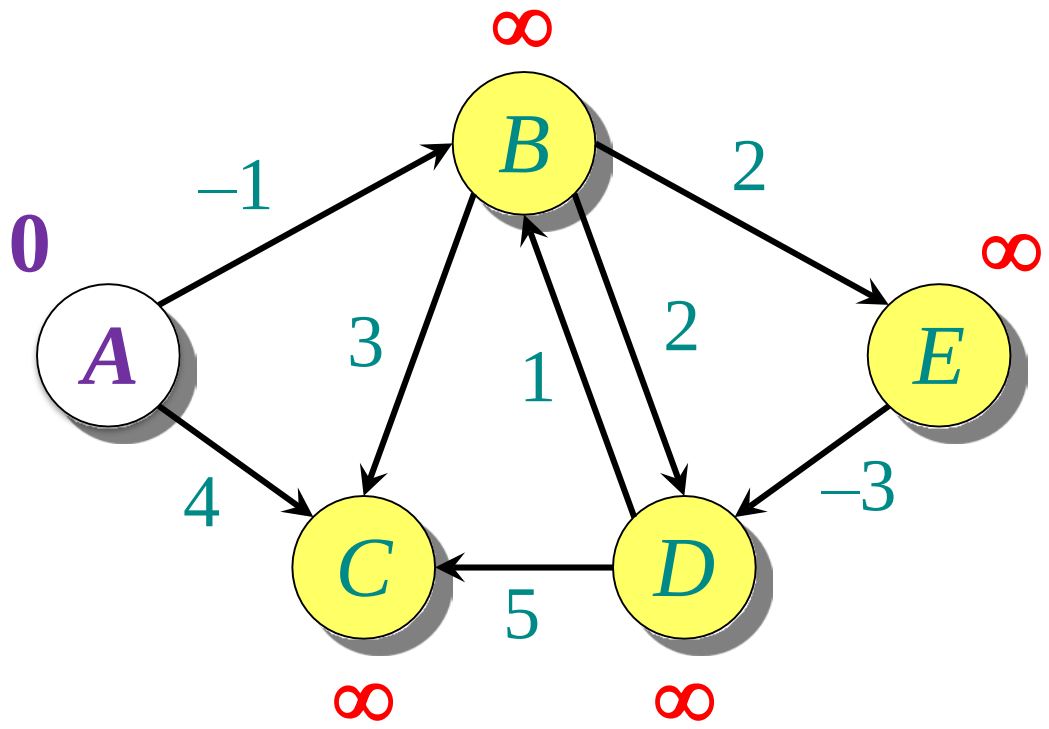
Example of Bellman-Ford



Initialization.

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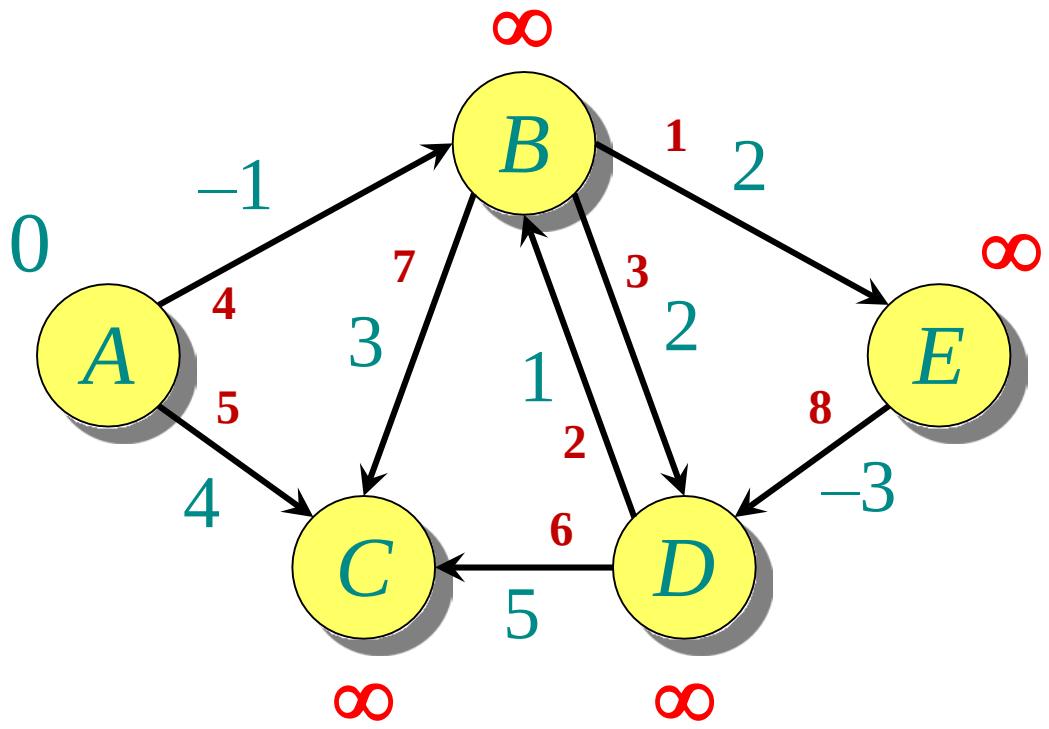
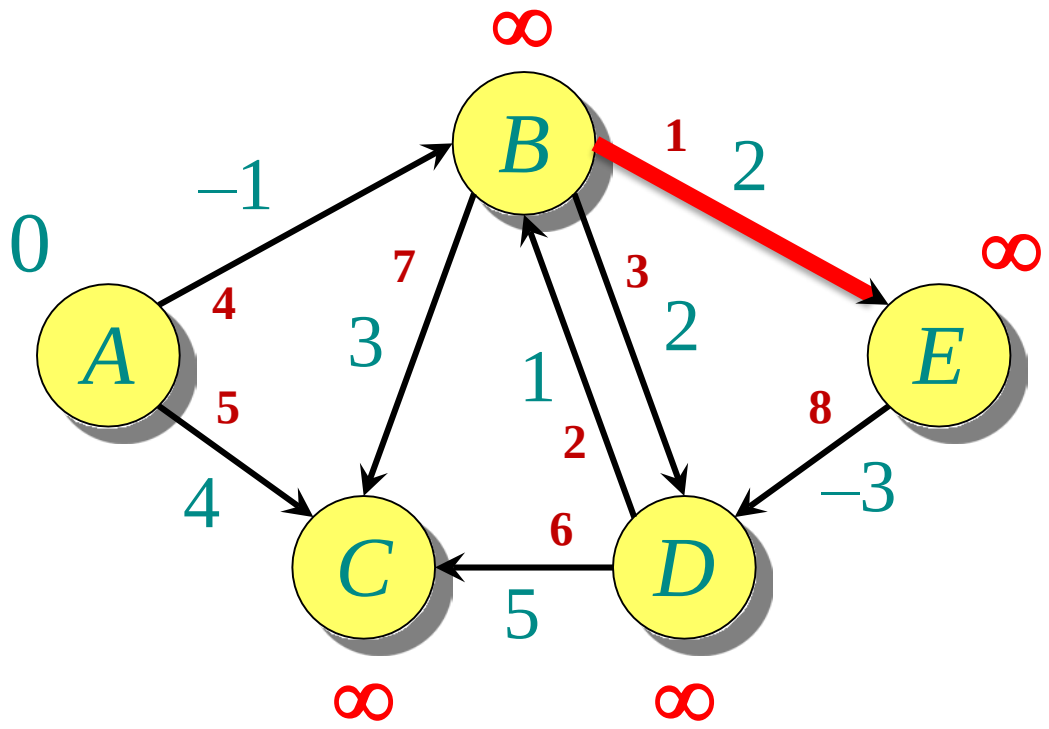
Example of Bellman-Ford

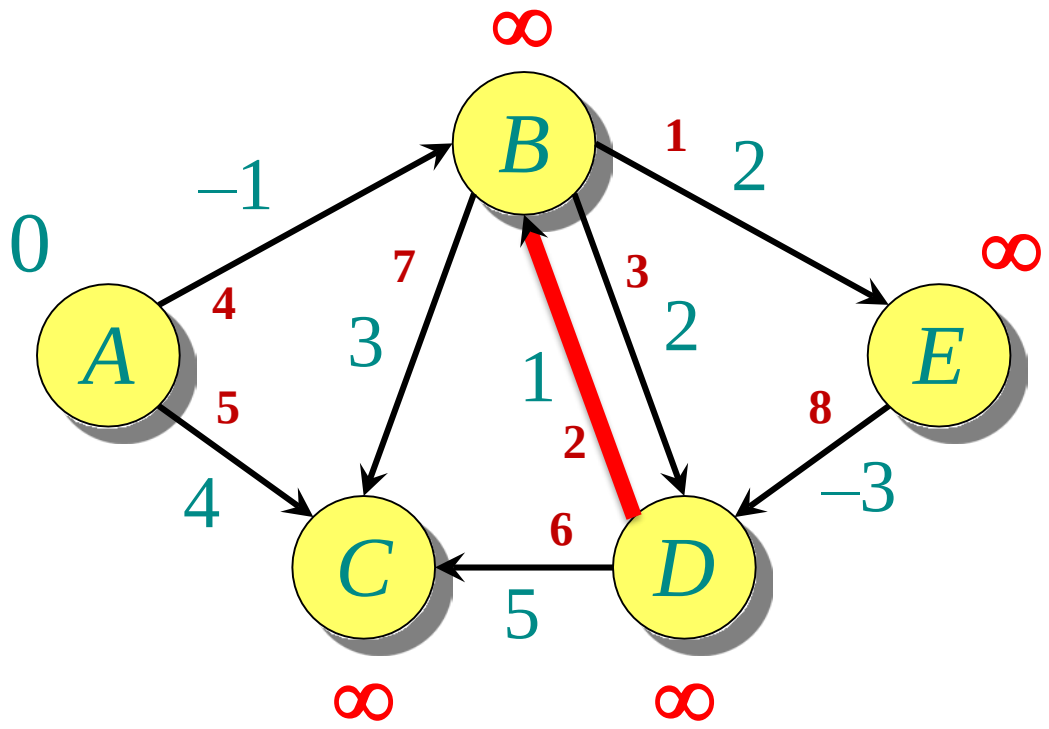


Note: Red number with edges is showing order of edge relation

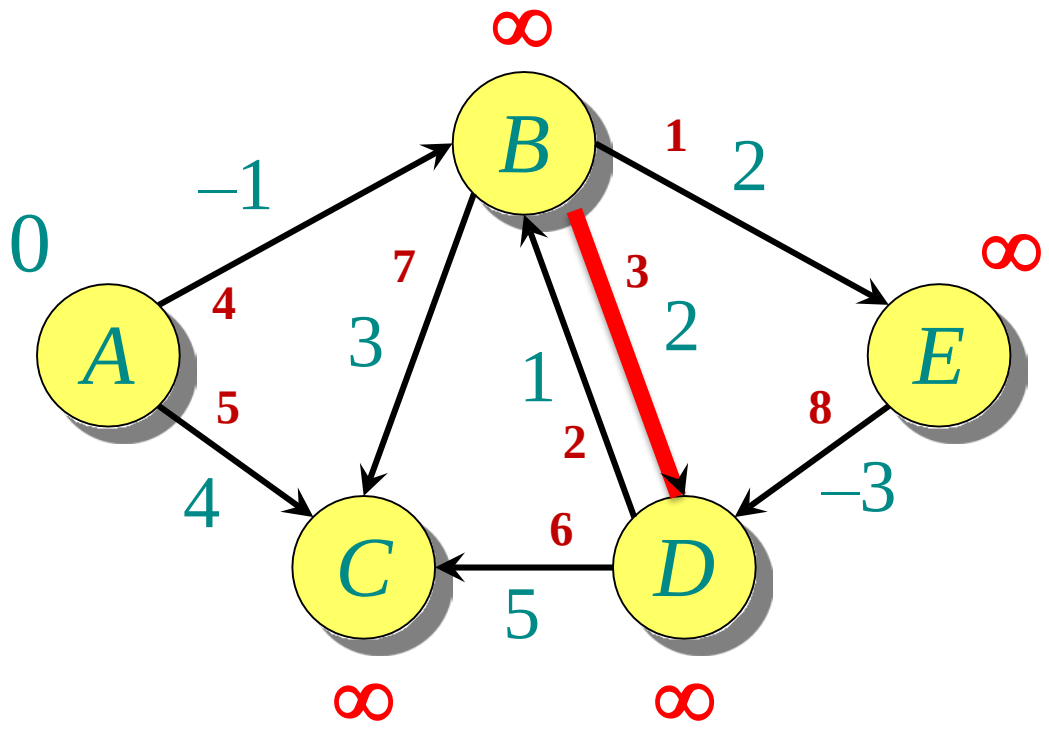
Example of Bellman-Ford



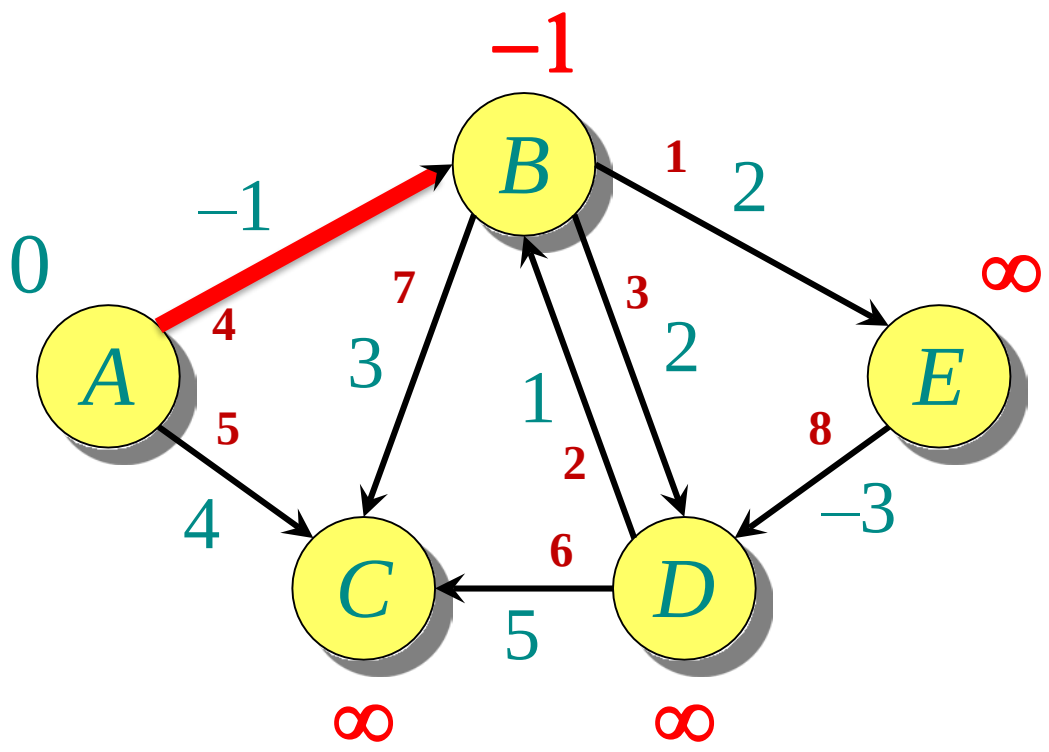
Example of Bellman-Ford



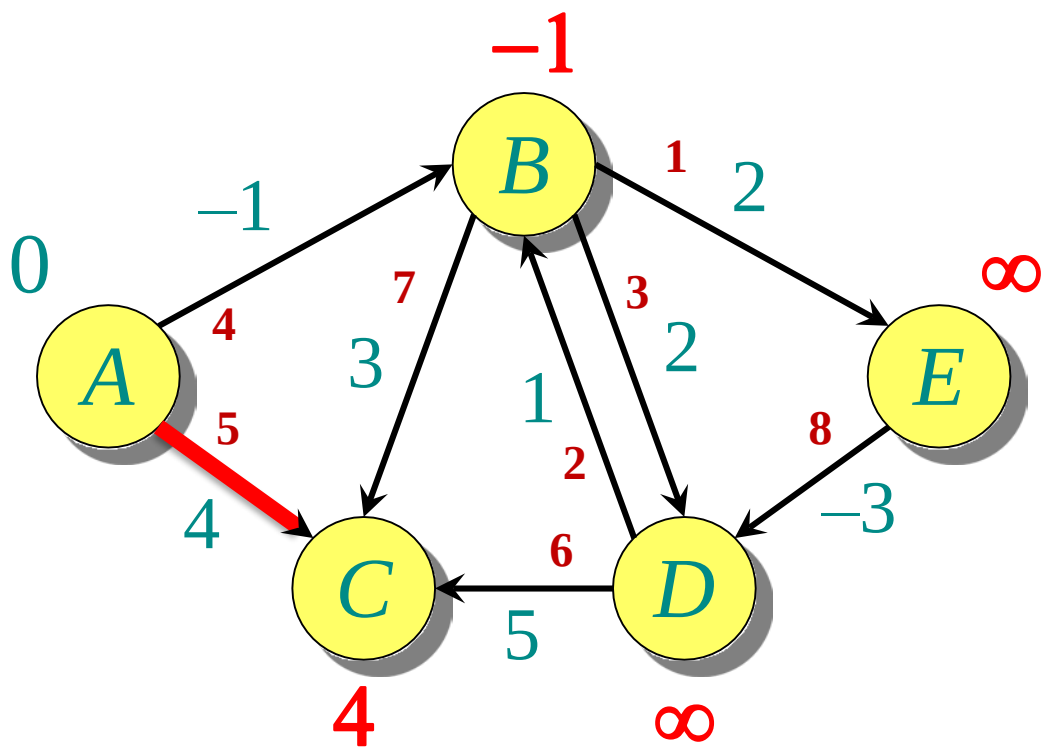
Example of Bellman-Ford



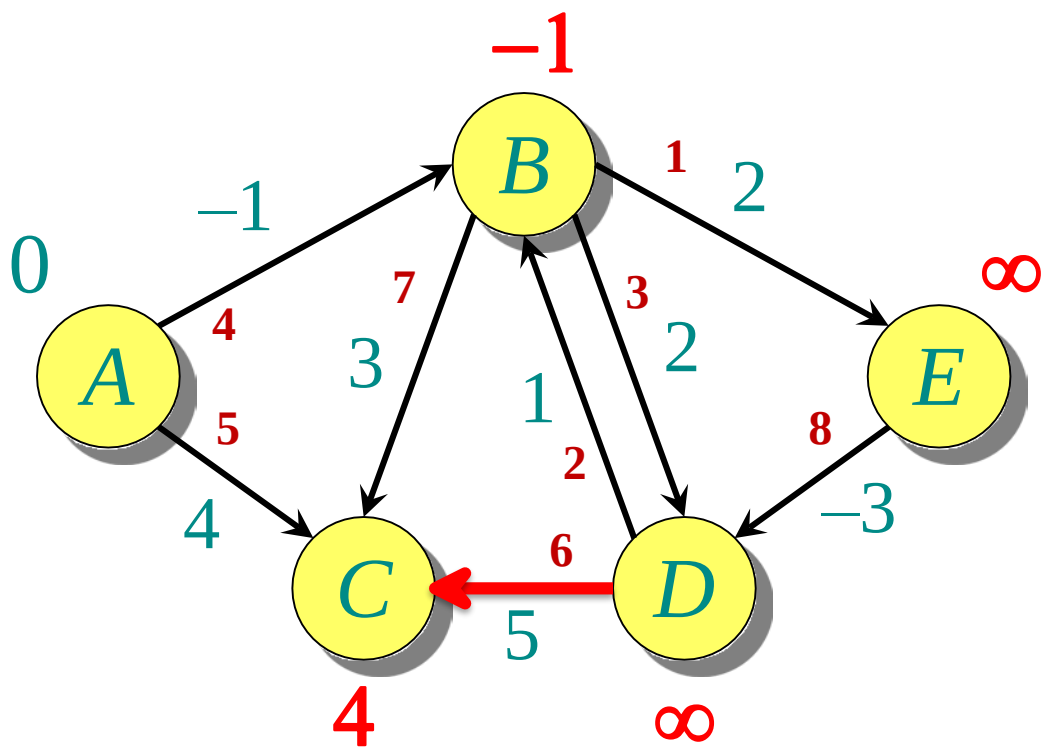
Example of Bellman-Ford



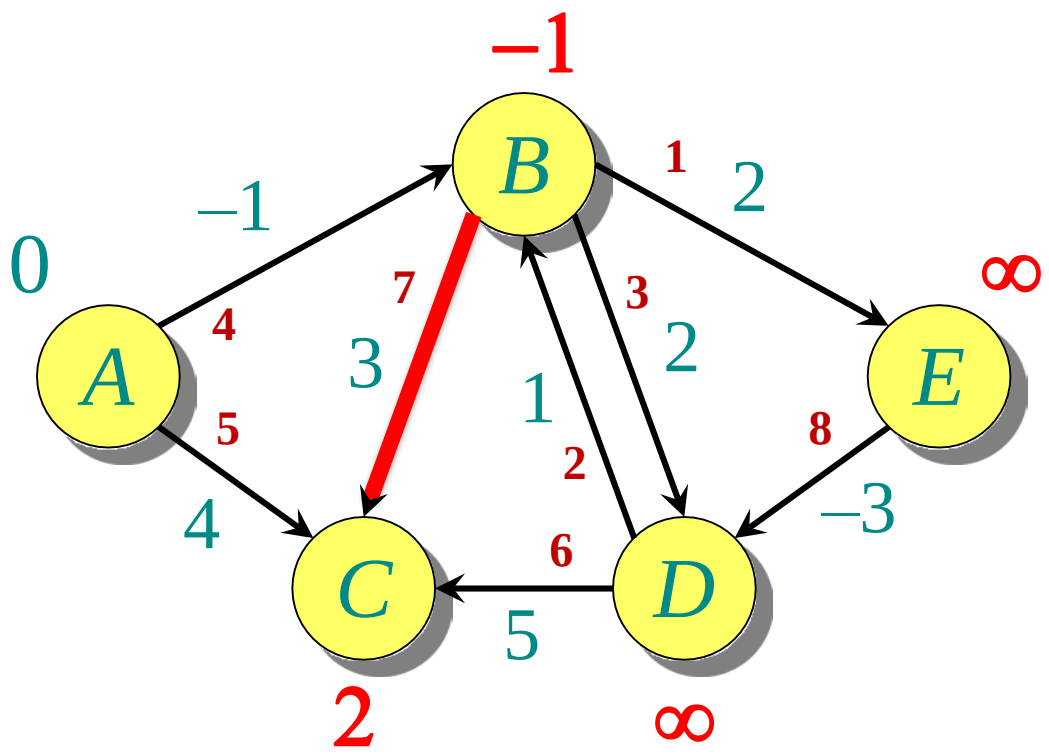
Example of Bellman-Ford



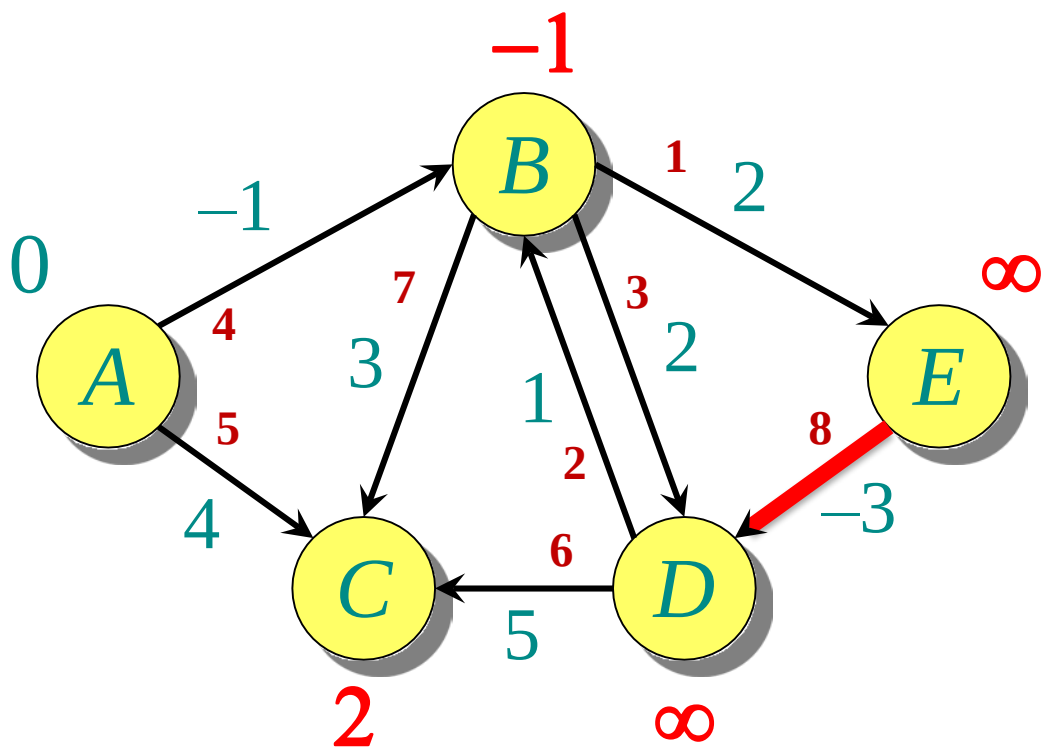
Example of Bellman-Ford



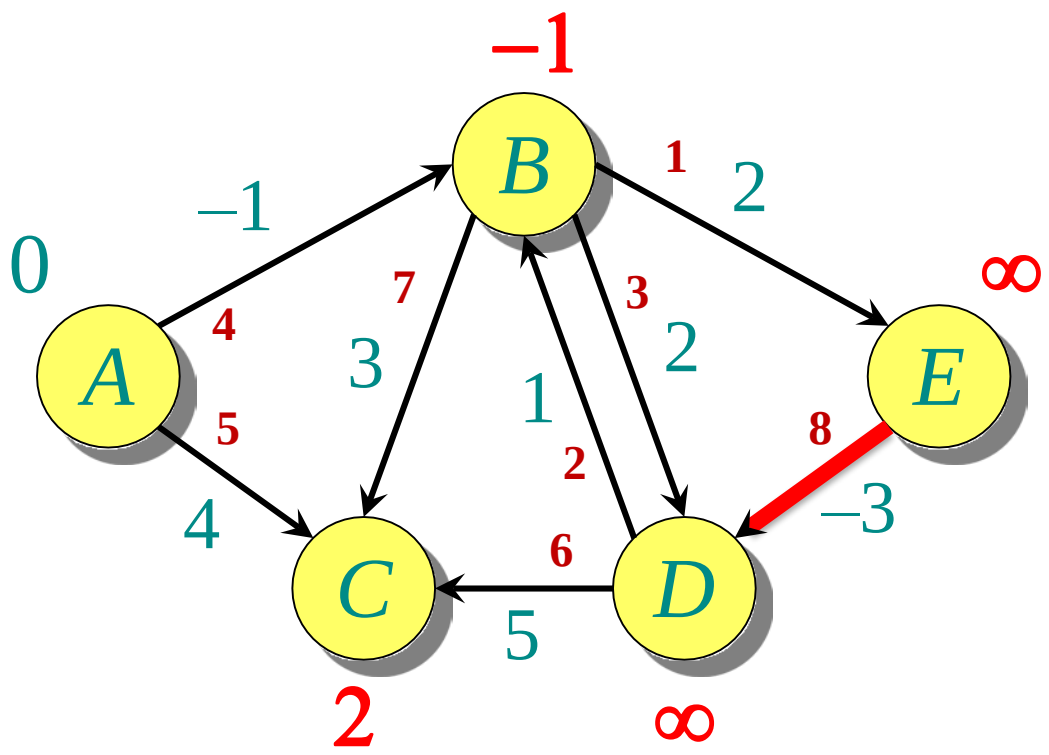
Example of Bellman-Ford



Example of Bellman-Ford

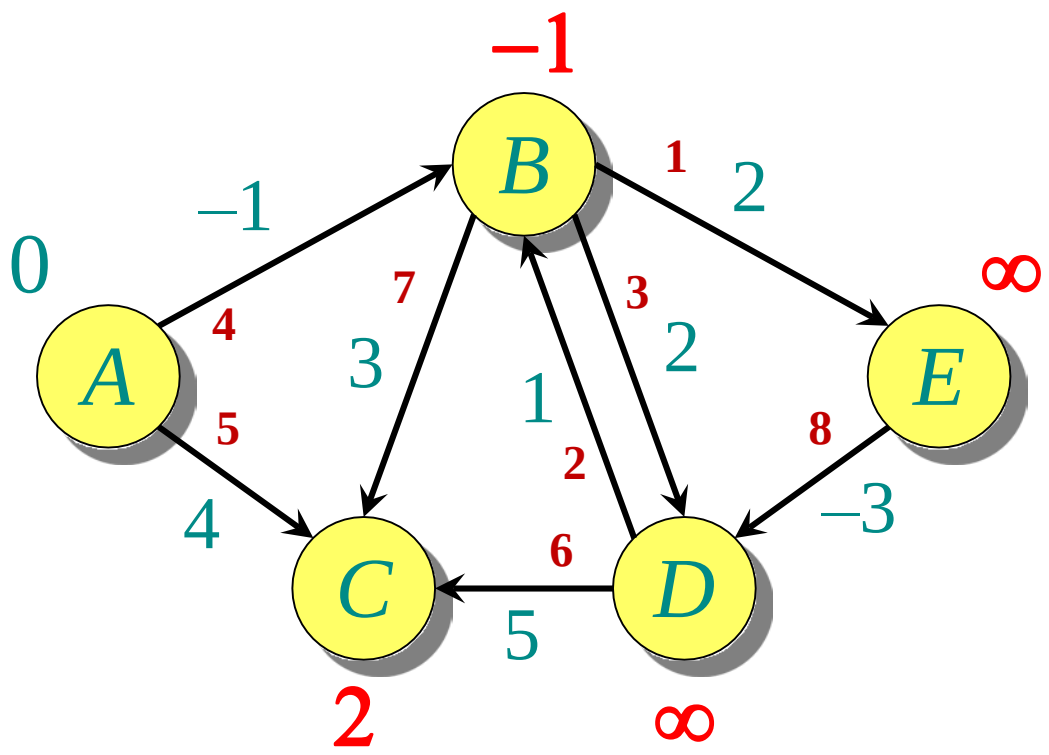


Example of Bellman-Ford



End of Pass 1

Example of Bellman-Ford



End of Pass 1

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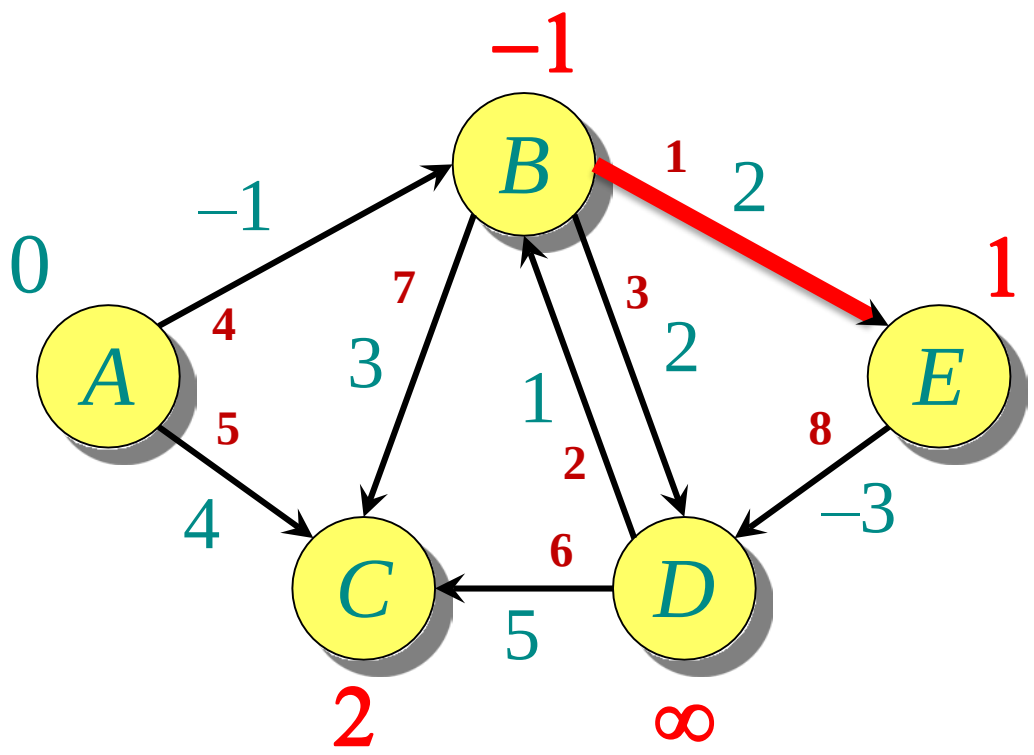
Example of Bellman-Ford-Pass2



Note: Red number with edges is showing order of edge relation

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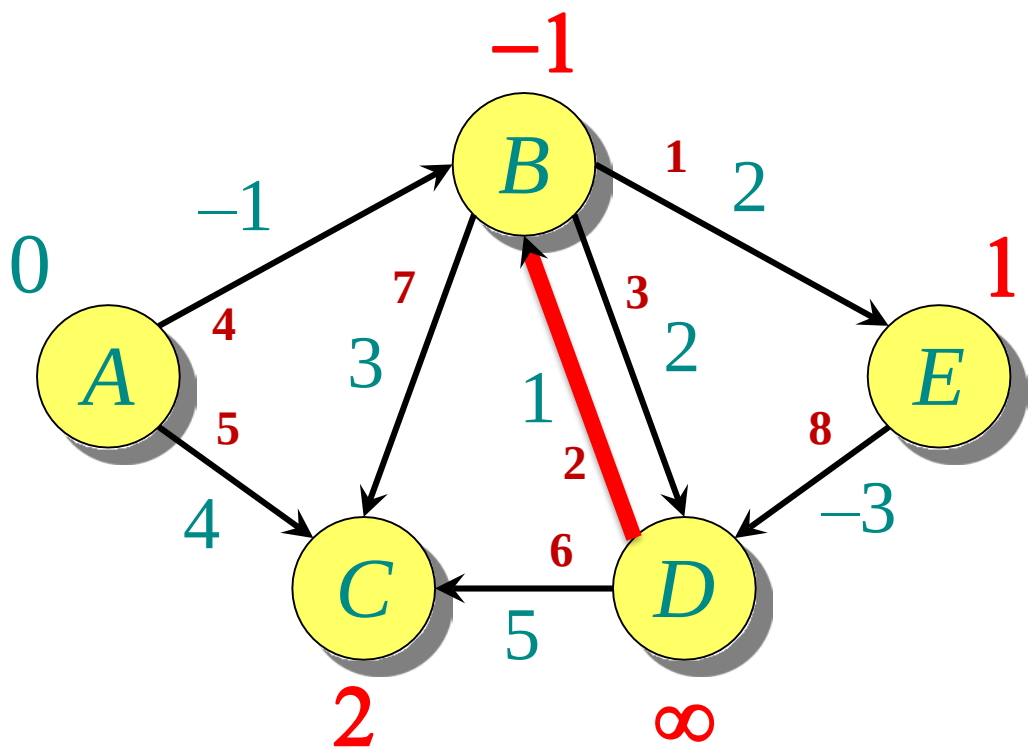
Example of Bellman-Ford



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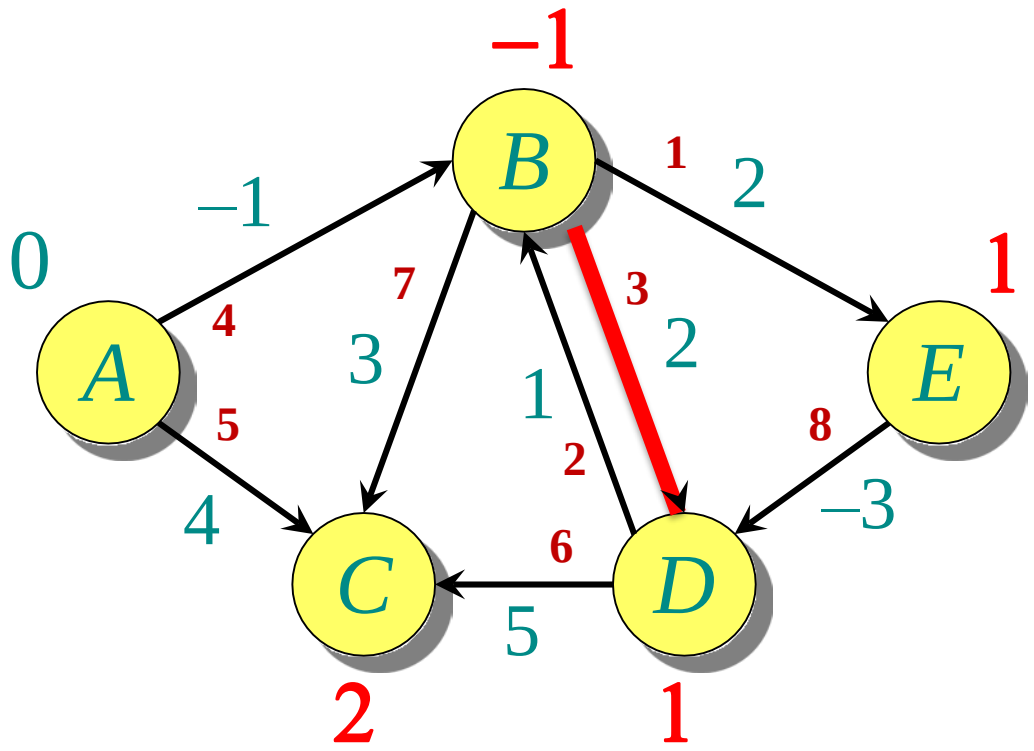
Example of Bellman-Ford



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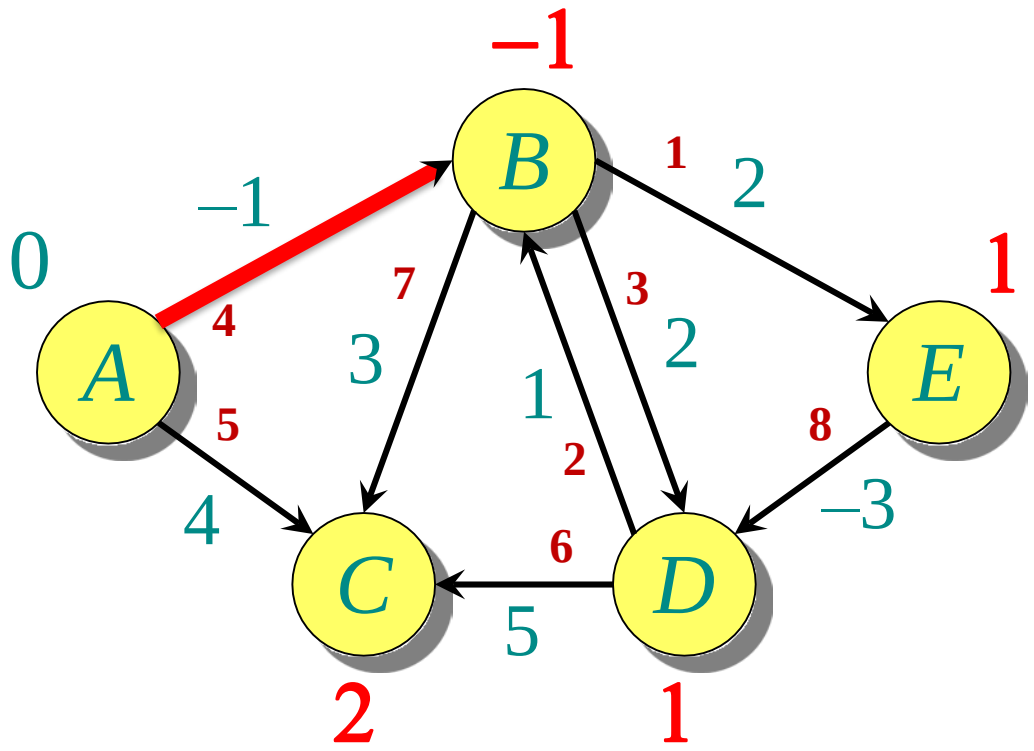
Example of Bellman-Ford



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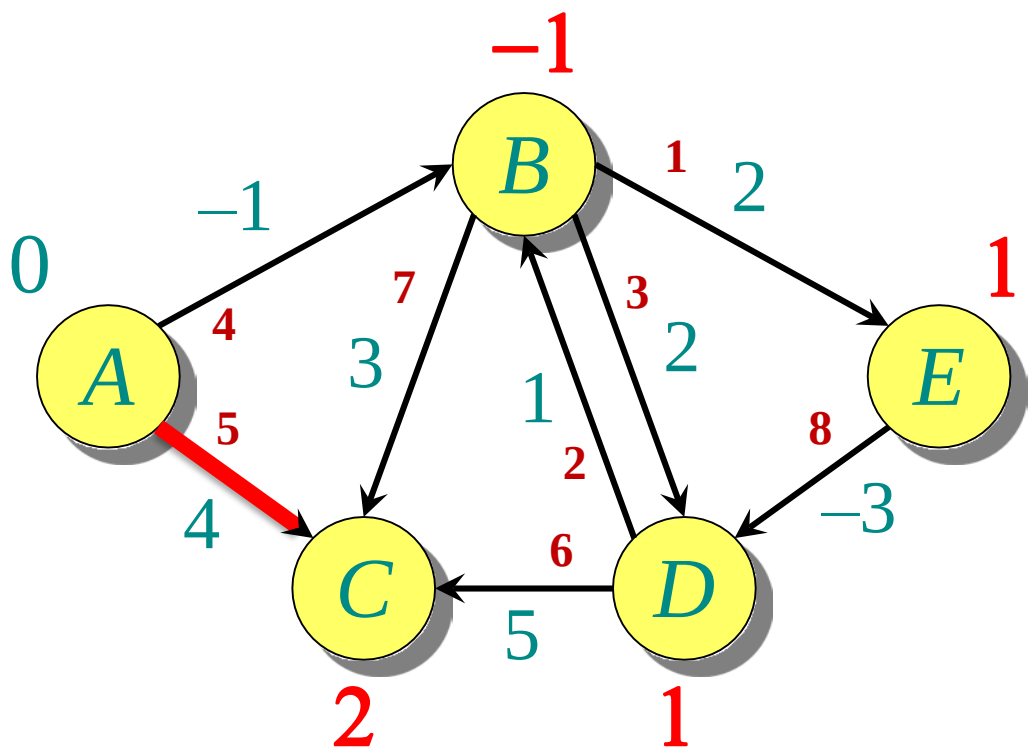
Example of Bellman-Ford



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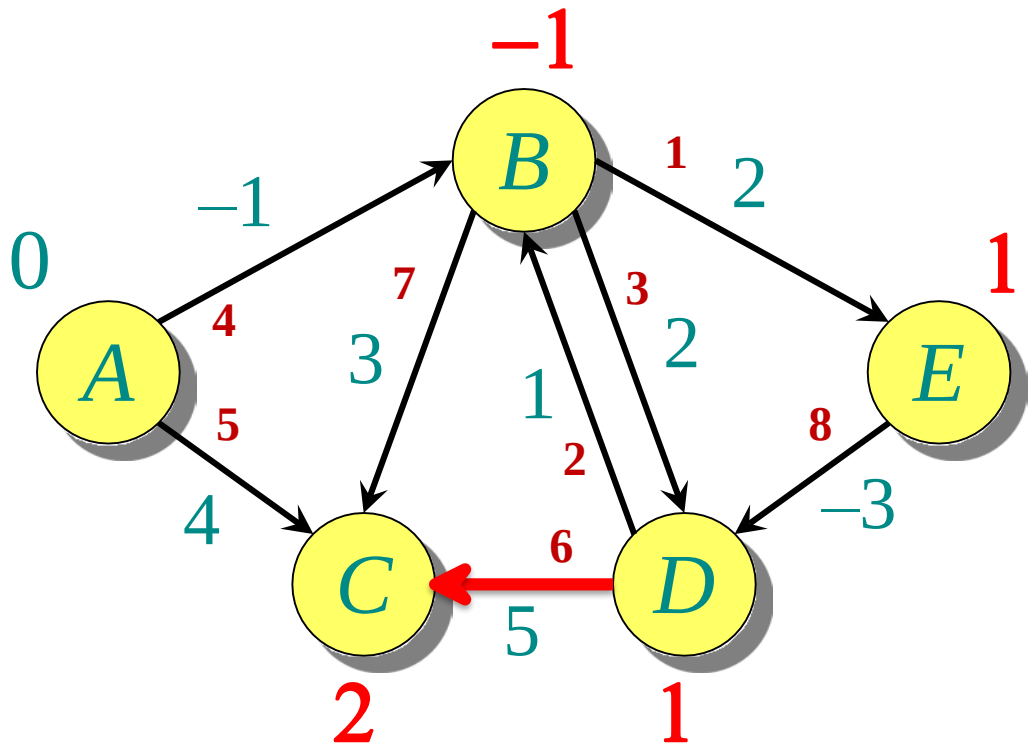
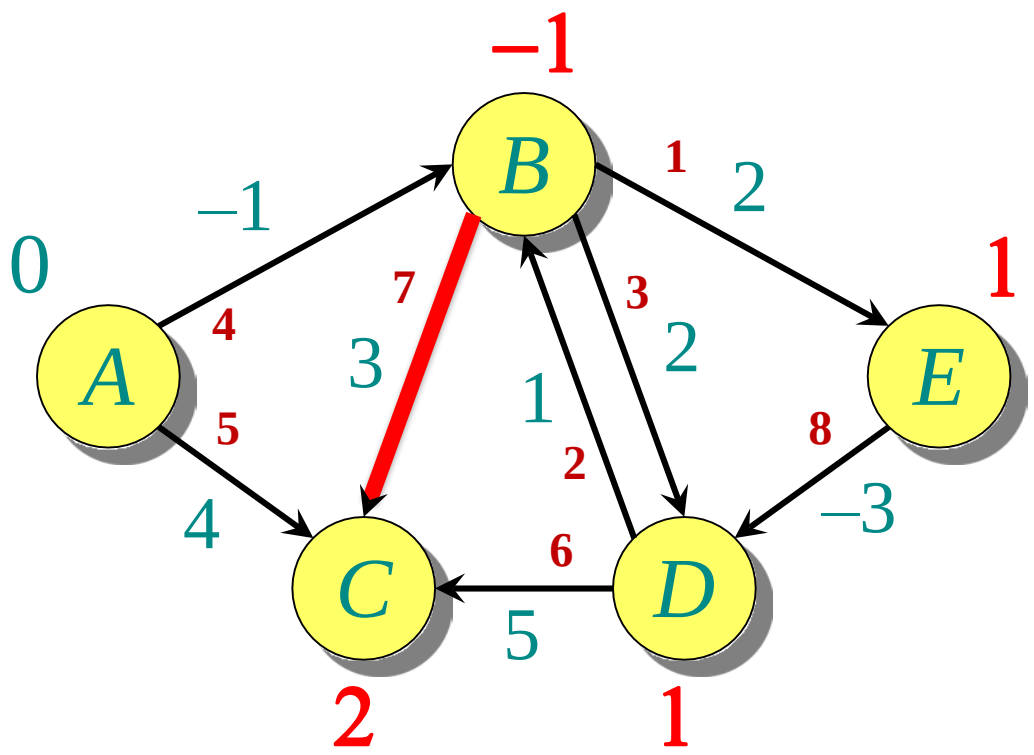
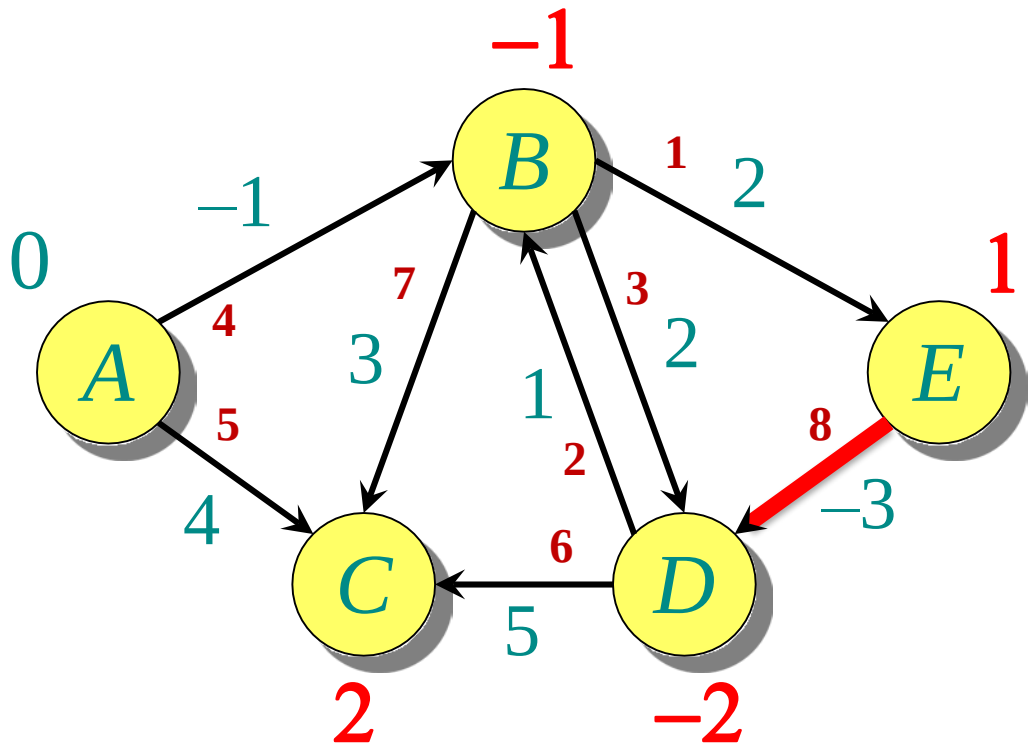
Example of Bellman-Ford



Example of Bellman-Ford



Example of Bellman-Ford



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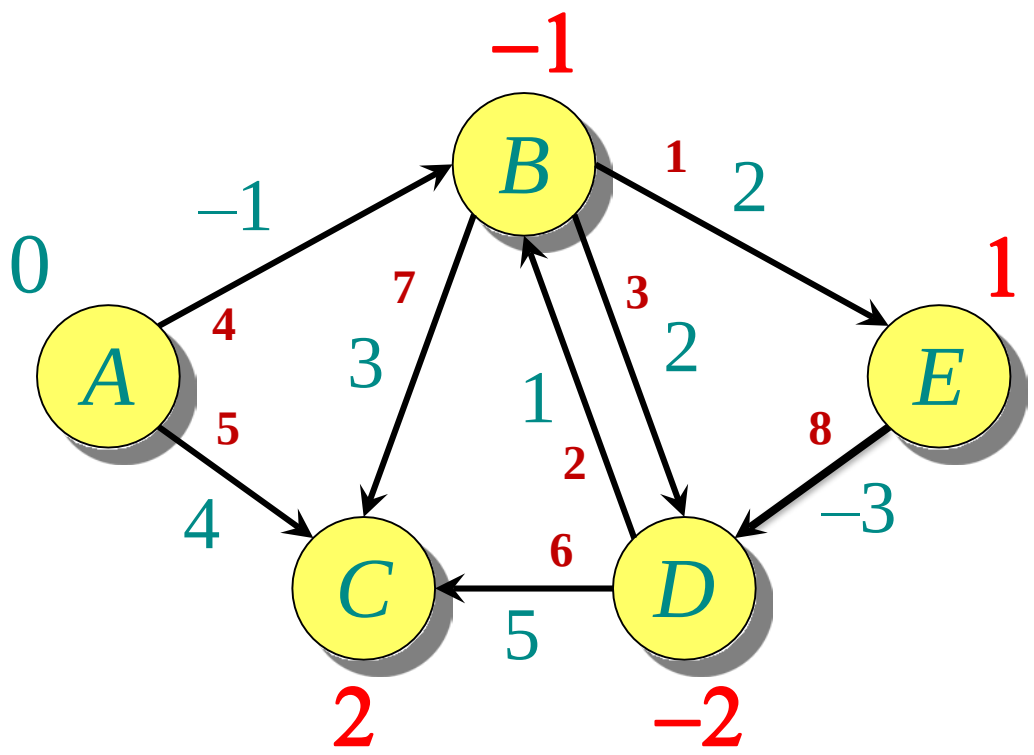
Example of Bellman-Ford



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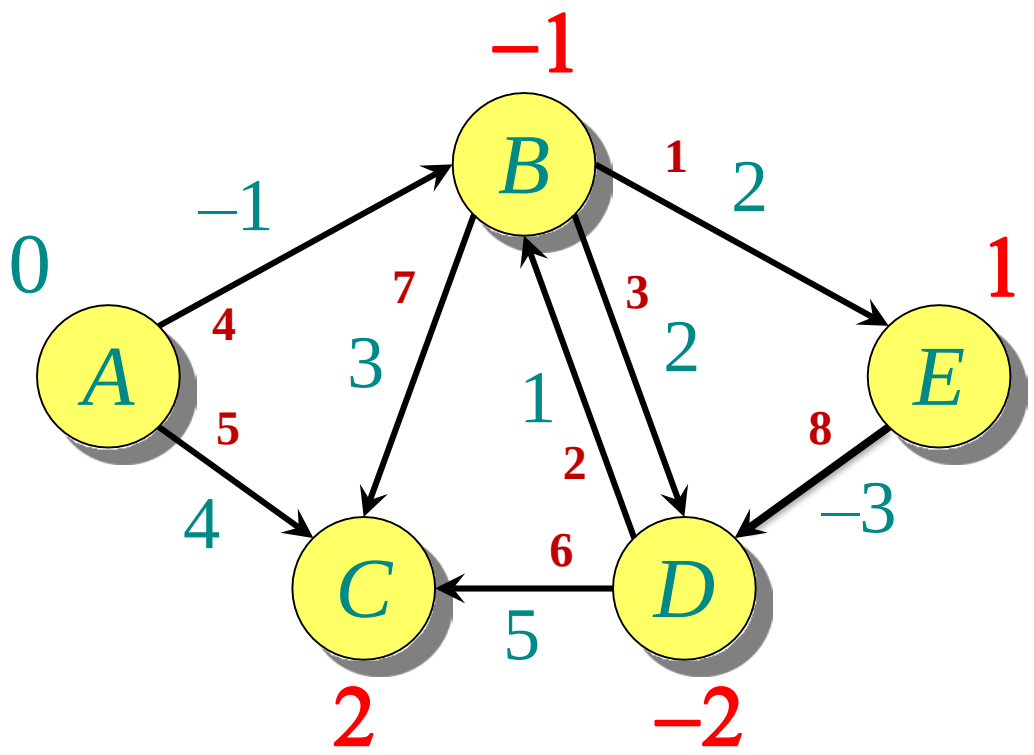
Example of Bellman-Ford



End of pass 2 (and 3 and 4).

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